Candlestick Pattern Recognition:

A Viable Algorithmic Technique for Amateur TradingView Users?

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Introduction

Algorithmic trading has become an increasingly accessible tool for retail investors, yet the gap between accessibility and profitability remains poorly understood. While institutional traders leverage sophisticated machine learning models, proprietary data, and high speed internet connections to inform trade decisions, retail traders are largely limited to publicly available or free platforms. Despite the growing prevalence of algorithmic trading among retail investors, limited research has examined the effectiveness of multi-indicator strategies built within the constraints of such platforms, such as TradingView. Technical analysis indicators, including candlestick patterns, support resistance levels, and moving average, have each exhibited individual merit in the literature. Nevertheless, few studies have examined their combined effectiveness within a single automated strategy designed specifically for retail traders. The present study addresses this gap by investigating to what extent a candlestick pattern recognition, sentiment-proxy indicator, and support resistance algorithm provides an effective trading strategy for retail investors on the AAPL daily time frame.

Literature Review

Algorithmic trading has become one of the most widely used applications in quantitative finance (Oyeniyi et al., 2024). In algorithmic trading, automated systems control trade entries and exits without human intervention (Treleaven et al., 2013). Technical indicators are core to such algorithmic trading strategies (Treleaven et al., 2013). Whereas fundamental traders make trades based on factors in corporate events or a company's subjective performance, algorithmic traders rely on numerical figures to predict price movement (Zucchi, 2021; Aycel & Santur, 2022). Fundamental analysis may examine a company's long term statistics such as liabilities and earnings, or even environmental or political factors. On the other hand, technical analysis

involves price-based indicators. This process done by algorithmic traders is better known as technical analysis, in which technical indicators calculate metrics such as price, trade volume, or momentum (Mayall, 2006). These indicators then supply algorithmic models that carry out the trades.

Many researchers are examining how technical indicators can enhance the accuracy of price-movement prediction (Billah et al., 2024). One of the most prevalent types of indicator is the moving average (MA), including the simple moving average (SMA), exponential moving average (EMA) (Billah et al., 2024). Most commonly, these MA indicators are functions of price where each new value is calculated by averaging prior price movement over a given time frame (Ahmad et al., 2017). For example, the SMA of a security is calculated by this formula:

(1)

$$SMA = \frac{1}{n}\sum_{i=1}^{n} A_i$$

where:

$A = average\ in\ period\ n$

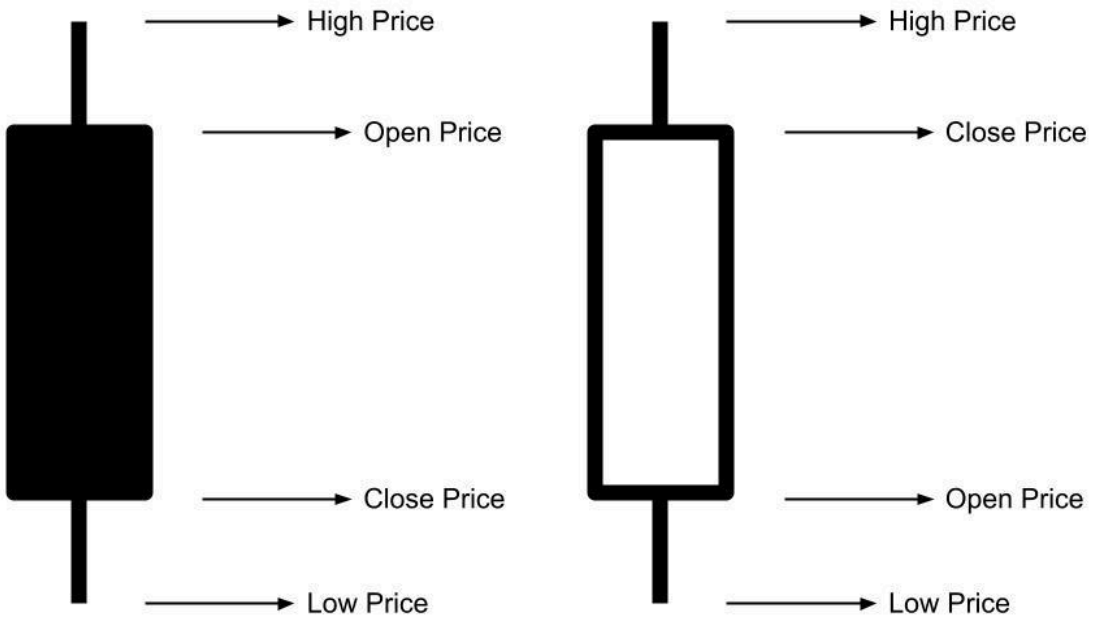
$n = number\ of\ periods$

MA strategies use daily signals to make trades based on price. If the strategy trades too frequently then it may be vulnerable to high transaction costs that weigh down or negate the profit margins. Consequently, longer lag lengths (longer length of a period) are common in MA strategies, since longer MA's result in longer average holding days. Shorter lags like MA-10 are favorable over longer lag strategies, however (Ahmad et al., 2017). Often, these indicators are illustrated on a financial chart representing financial data.

Both fundamental and algorithmic trading rely on heuristic decision making (Romo et al., 2025). Heuristics, as they relate to finance, are assumptions made based on logical explanations for market behavior.

Popular with both fundamental and algorithmic traders, the candlestick chart is a financial chart that shows price movement along with the low and high for a time period (Aycel & Santur, 2022). Each period corresponds on the chart to a vertical rectangle with wicks on both ends, hence the visual similarity to a candlestick. Here, the wicks tell the low and high prices for a period. Example:

Figure 1. *Understanding the candlestick chart.*



Note. From Tradingview [Screenshot], by TradingView, [2026] (https://www.tradingview.com).

The nature of the candlestick chart in showing 4 data points (high, low, open, and close) of the price for each period makes it especially useful for algorithmic traders, providing multiple price data points per period to evaluate within technical indicators (Ho & Huang, 2021). An SMA graphed on a candlestick chart looks like this inside the TradingView application:

Figure 2. *SMA and candlestick chart for Tesla stock, January-March 2024.*



Note. From Tradingview [Screenshot], by TradingView, [2026] (https://www.tradingview.com).

Some studies find that patterns on candlestick charts can predict price movement independently from technical indicators. There are many candlestick patterns that each help tell bullish or bearish trends, reversals or continuations: Doji, opening and closing prices are very close to each other; Doji star, an undecided candle whose lower and upper wicks are equal and

the opening and closing values are almost equal; Dragonfly Doji is a bullish signal that occurs when both the open and closed price is relatively close to the high price of a day; Gravestone Doji, the bearish counterpart to the Dragonfly Doji; among others. Whereas moving averages and other adjacent indicators may have issues with lag, candlestick charts reflect the price-based psychology of bear and bull investors (Aycel & Santur, 2022). According to Aycel & Santur, 2022, candlestick charts can be used in almost any period. The study revealed that algorithmic trading framed solely from candlestick patterns can find success. Despite only 21 of the most popular patterns being used in this study, the profit/loss ratio could be above 90%.

Support Resistance (SR) is one of the most crucial tools in technical analysis due to its broad assessment of price movements, numerical rules, order flow, and market psychology (Chan, et al., 2022). Therefore, SR may be applicable to many types of strategies. SR is an uncommon topic in scientific studies since it can depend on personal perception. Since it may depend on public perception, it can be more difficult to study. SR may have many interpretations: it is used differently across timeframes, a level significant on a weekly chart may be irrelevant on a 5-minute chart; SR can be either diagonal or horizontal, spurring debates about its efficacy; one price level can simultaneously act as support for one trader and resistance for another depending on their entry point; institutional and retail traders often draw SR at different levels due to different data access and charting tools. Nonetheless, research finds it is likely an effective indicator for algorithmic trading. SR may be identified by finding Swing High Swing Low endpoints algorithmically (Chan, et al., 2022).

These fundamental strategies preside sentiment indicators alongside basic technical indicators. In this difference, researchers have attempted to fuse sentiment analysis with algorithmic trading strategies in order to bridge the gap of knowledge provided by qualitative

and quantifiable metrics pertaining to financial securities (Gómez-Martínez et al., 2025). The appeal of a potential fusion can come from the efficient-market hypothesis, postulating that the prices of securities always reflect all currently available information, thus being an indicator of intrinsic value (Mayall, 2006). This contextualizes the benefits of sentiment analysis illustrated in Gómez-Martínez et al., 2025. Behavioral finance demonstrates that emotions like fear, greed, and overconfidence can influence price, supporting the logic in using sentiment analysis within an algorithmic strategy. This effect stems from collective investor behavior, which is measurable with sentiment indicators (Gómez-Martínez et al., 2025). Few studies have incorporated sentiment analysis into technical analysis with traditional algorithmic trading strategies, however there may be great benefit in doing so, especially for candlestick pattern strategies, (Ho & Huang, 2021; Gómez-Martínez et al., 2025). Gómez-Martínez et al., 2025, find how key performance indicators (KPIs) are efficacious to measure the performance of an algorithm. In their testing, the researchers find that their sentiment-augmented strategy exhibits greater risk-adjusted performance, stability, and capital efficiency compared to their AI-only approach. Across a variety of KPIs, sentiment indicators offer value to algorithmic trading strategies, although not necessarily providing superior absolute net profit. Aycel & Santur, 2022, found high levels of profitability (90%) when testing a candlestick pattern recognition strategy without sentiment component, nonetheless Ho & Huang, 2021, demonstrate that candlestick pattern recognition strategies can benefit from sentiment analysis. For instance, when testing a candlestick pattern recognition and sentiment indicator based strategy they garnered, at the highest, 75.48% prediction accuracy for AAPL and, at the lowest, 51.26% for IBM. Stocks AAPL, TSLA, IBM, AMZN, and GOOG were selected based on their market capitalization. The researchers in this study integrated stock sentiment analysis and its candlestick chart to estimate

the stock's price movement over the 4-, 5-, 8-, and 10-day time frames. Predictions overall favored longer periods of time compared to shorter periods of time. Apple stock (AAPL) is an especially relevant security for many traders, including retail investors, because of its high market capitalization, past high performance, and low volatility compared to similar securities as given by its beta (β) of 1.11, as well as its broad familiarity among retail investors. This beta value being slightly greater than 1.00 means AAPL volatility moves mostly in line with the market and is far lower than other high performing tech stocks such as Nvidia (NVDA) with a beta of 2.34.

Key performance indicators (KPIs) measure the performance of an algorithm. Some prevalent KPIs are Net Profit, Gross Profit, Profit Factor (gross profit to gross loss ratio), Sharpe ratio, Annualised ROI, Net P&L over Drawdown, Mathematical Expectation, and Volatility. Across a variety of KPIs, sentiment indicators offer value to algorithmic trading strategies, although not necessarily providing superior absolute net profit (Gómez-Martínez et al., 2025). Trade strategies can differ among traders based on portfolio, capital, and other opinionated factors (Mayall, 2006). For example, institutional traders buy and sell securities for accounts that are managed by a group or institution, whereas retail traders manage the buying and selling of securities in their individual trading accounts (Zucchi, 2021). Characterized by their distinct differences in resources, methodology, and time frame, retail and institutional traders buy and sell securities in unique ways. In the past, institutional traders had major advantages over retail traders. But, in recent years the gap has narrowed due to the newfound widespread access to financial data. Yet, institutions still have advantages such as access to more securities and the ability to negotiate when needed. Because of this, institutional traders often make trades that can have larger impacts on the market. Retail traders have minimal access to IPOs and overall have

higher costs to make trades (Zucchi, 2021). Ultimately, less research has been done on trade strategies for retail traders, hence the gap in the literature which will be examined in the present paper.

Stop loss is a system that exits a trade before profitability is lost. Vaikole & Yash, 2022, find that stop loss is important for managing risk and supplementing profitability. An average true range (ATR) stop can be especially effective. A trailing stop loss automatically calculates new stop loss values according to price or other factors. ATR is a common type of trailing stop. In the past, this strategy had been done only by institutional traders or investment firms, however with the onset of algorithmic trading this strategy is markedly more accessible to both professionals and new traders.

Method

The present study follows a quantitative design based on computation and backtesting performance of financial trade decisions. It also follows ethical guidelines as there is no plagiarism and no human subjects involved. These trade decisions were generated by an algorithmic model using publicly available Apple Inc. (AAPL) stock-price data. AAPL was chosen for its relative price stability, reflected by its five-year monthly beta (β) of 1.11 (Yahoo Finance, n.d.), indicating modest market sensitivity compared to other, more volatile, high performing securities. This study was conducted with the TradingView financial analysis platform and by using its integrated Pine Script® (version 6) programming language to construct the trading algorithms. This approach is well supported by the literature: algorithms in trading minimize error in human judgement, therefore reducing potential for incorrect trade decisions (Mayall, 2006). This is outlined as one of four subcategories of technical analysis used by individual share traders. The present trading strategy was constructed according to the literature

on the use cases of candlestick charts, support and resistance levels, technical and sentiment analysis, and moving average trading strategies, as well as being informed by the idea of price momentum: high performing stocks will continue to excel against low performing stocks (Treleaven et al., 2013). Particularly, the present algorithm chiefly included sentiment-proxy indicators attempting to replicate the CNN Fear and Greed Index within a candlestick-pattern recognition trading strategy that identifies 4 bullish candles: hammer, three white soldiers, inverted hammer, bullish engulfing; 3 bearish candles: three black crows, shooting star, bearish engulfing; 2 consolidation candles: doji, spinning top. The CNN Fear and Greed Index proxy was built based on the logic for the 7 individual indicators on its webpage (CNN, n.d.). This included several MA components such as a VIX 50-day MA, for instance. Research by Ho & Huang, 2021, found hybrid strategies, such as in the present study, beats algorithms built on candlestick charts or sentiment analysis alone, hence the choice to include both technical analysis and sentiment analysis based indicators. More research finds that sentiment-augmented strategies can exhibit greater risk-adjusted performance, as well (Gómez-Martínez et al., 2025). The present strategy was backtested and evaluated based on profitability and other aspects inside of TradingView. Profitability and risk-adjusted performance were calculated using the built in "Strategy Report" feature that comes after an automated backtest on all free financial data accessible in TradingView. All price movements and subsequent trade decisions were represented on candlestick charts, with the primary chart displaying the "Tab" for the current security being traded. Backtests process prior data to replicate the algorithm's trade decisions had it begun trading at the beginning of the backtested period. Accordingly, backtests are crucial to evaluate the effectiveness of an algorithm. However, Gómez-Martínez et al., 2025, postulate that algorithmic overfitting is a danger when backtesting. Overfitting occurs when an algorithm

greatly succeeds in backtesting but falls short in live market results. Yet, they find that sentiment indicators could combat overfitting, which influenced the inclusion of sentiment analysis in the present strategy. On top of the convenience of the experiment being run inside TradingView, it offers unique advantages for this study. Note, however, that Pine Script does not natively support calls from external sources, which is why the present study uses sentiment-proxy indicators rather than capturing the true CNN Fear and Greed Index value. The use of TradingView's free data will increase the applicability of this study's findings for retail or amateur traders. Retail traders have less capital and data access than institutions and hedge funds (Zucchi, 2021). Much of the current literature pertains to institutional traders in general. Moreover, common institutional trading methods such as high frequency trading (HFT) exploit ultra-low-latency technologies to gain advantage over competitors by the order of millisecond level trade decisions (Oyeniyi et al., 2024). As such, HFT and other institutional trading methods are essentially unavailable for retail traders, which further supports the use of TradingView — a popular, publicly available application — in the present study. The present independent variable is the algorithmic trading strategy, its calculations and logic, with the dependent variable being KPIs including profitability. Mediating variables include the sequence of trade positions, i.e., long, short, or hold, which are themselves dictated by the algorithmic strategy. As for stop loss, the present study used an ATR based trailing stop because of the benefits seen in Vaikole & Yash, 2022. ATR is inherently adaptive to the current conditions of the given security, helping ensure maximization of trade profitability. A fixed percentage stop, on the other hand, may introduce unnecessary risk in certain conditions (Vaikole & Yash, 2022).

In summary, the available literature guided the addition of each component comprising the design of the present trading strategy and guided the subsequent performance analysis.

Indicators were vetted based on past performance in the research and synthesized into one system, utilizing each in a logical manner suitable for a retail trader using TradingView.

Results

The backtesting simulation carried out on the AAPL daily time frame from 2007-2020 produced 86 total trades, 19 of which were profitable. Trades were distributed relatively evenly throughout the test period. See Table 1 for Key performance indicators from TradingView.

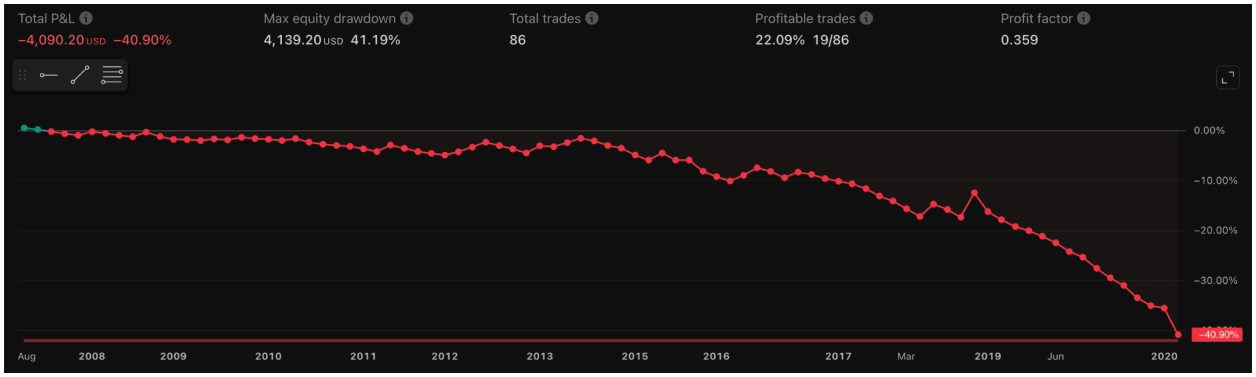
Table 1. *Backtesting KPIs for AAPL Daily Strategy.*

Total Trades	86
Win Rate	22.09%
Profit Factor	0.359
Gross Profit	22.94%
Net Profit	-40.90%
Buy & Hold Return	1,219.34%

As per Table 1, the present strategy executed 86 Total Trades with a Win Rate of 22.09%. Furthermore, it produced a Profit Factor < 1, meaning more money was lost than gained. Additionally, negative Net Profit of approximately double the Gross Profit is observed,

displaying the algorithm's overall failure to sustain consistent profitability. The Buy & Hold Return, in comparison, achieved great profitability.

Figure 3. *Equity Curve for the present strategy performance.*



Note. From Tradingview [Screenshot], by TradingView, [2026] (https://www.tradingview.com).

As shown in Figure 3, the Equity Curve (a graph showing the portfolio's change in value over time) demonstrates a consistent downward trend throughout the test period (2007-2020), with no sustained periods of profitability, suggesting the strategy lacks a consistent edge across varying market conditions.

Risk analysis was conducted for this backtest. The strategy achieved an alpha (α) of -1,260.24%, indicating substantial underperformance relative to the market benchmark, as given by this formula:

(2)

$$\alpha = Strategy\ Return - Market\ Return$$

Moreover, the Sharpe ratio and Sortino ratio were calculated by the included TradingView function, being -0.349 and -0.358, respectively. Both ratios being < 1 indicate that undertaken risk each trade did not justify the potential returns.

Discussion

This study sought to determine to what extent a candlestick pattern recognition, sentiment-proxy indicator, and support and resistance algorithm produces an effective strategy for retail trading on the AAPL daily time frame. The backtest results ultimately indicate that the strategy was not effective, producing a net return of -40.90% and an alpha of -1,260.24% against a buy and hold return of 1,219.34%. To better understand why the strategy underperformed, the following section analyzes each of the three primary indicators in use. The candlestick pattern recognition system likely fell short due to it not utilizing a machine learning backing to infer the recent trading patterns and correlate that with trade decisions. Research has found a candlestick pattern recognition system can be an effective strategy (Aycel & Santur, 2022). Where such successful strategies differ from the present algorithm is chiefly in the incorporation of a neural network that learns which candlestick situations and trade decisions generate profits. In the case of Aycel and Santur, their strategy involved ensemble learning, a machine learning technique combining two simultaneous neural models outputting statistical predictions. As for the sentiment-proxy indicator, this component of the strategy was reasonably effective. The literature has thoroughly shown the benefits of sentiment analysis in trade decision making. However, much less research has been conducted relating to incorporating sentiment indicators into algorithmic trading. One study from Ho & Huang, 2021, particularly implements sentiment

analysis into a candlestick based trading strategy, similar to what is done in the present study. Once again, the lack of a machine learning based system of candlestick pattern recognition is essentially what generates the observed unprofitability, rather than the sentiment analysis. Nevertheless, it is important to note that the present study did use a sentiment-proxy indicator, that is in this case, a reproduced indicator mimicking the CNN Fear and Greed Index. Naturally, there will be some residual between the true Fear and Greed Index value and the reproduced sentiment-proxy indicator. Still, the outputs produced by the sentiment proxy were relatively accurate to the CNN Fear and Greed Index and thus not likely to be the root cause of the unprofitability. Lastly for the three primary components, the support and resistance system had occasional problematic responses. The research is firm that support and resistance is a highly important indicator in technical analysis (Jireh Yi-Le Chan et al., 2022). However, several key challenges arise with integrating support and resistance into an automatically trading algorithm. Support and resistance levels are most commonly interpreted via a trader's intuition, and while this doesn't necessarily mean that support and resistance is ineffective in an algorithmic strategy, it does additionally highlight the benefit of machine learning for this purpose. A machine learning algorithm, such as using the ensemble learning technique from Aycel and Santur, could better infer support and resistance levels while referencing them in the trade decisions. A purely numerical algorithm without a hard-coded decision system would lack the ability to infer how to trade given the current support and resistance situation in the context of the other indicators. The present strategy did include hard-coded parameters to aid in using the support and resistance as a supplemental filter to the algorithm, and it identified relatively accurately current support and resistance levels. Yet, its benefit is unfortunately limited without the machine learning backing. These issues ultimately demonstrate the limitations of TradingView software. Being a free

resource with much financial data available, it is inherently an effective tool for amateur or retail traders. However, the Pine Script language lacks elements like pulling data from custom, external sources. Therefore, a Python coded machine learning strategy could be more effective for the present indicators in use. This limitation more so illustrates the challenge of coding a multi-indicator strategy in Pine Script rather than a lack in the logic behind the indicators used, as suggested by the profitability of similar strategies when coded in a more sophisticated software library tailored to the specific needs of the strategy. Regarding limitations in the results of the present study, one security (AAPL) and one timeframe (daily) are tested on. This could mean that there exist significantly different levels of profitability contingent on the security being traded. Therefore, the findings should not be generalized to other classes of stocks, sectors, or assets. Notwithstanding this, it is reasonable to expect that other high-performing, relatively low volatility stocks like AAPL will produce similar results. Another limitation comes from the use of the backtest. While backtests are critical for various financial trading studies, they do present some issues. As described in Gómez-Martínez et al., 2025, algorithms can overfit to the backtesting scenarios, producing different returns than expected when trading on a live market. In the case of the present study, there exists some potential that live market testing would generate greater returns. Moreover, backtesting does not account for transaction costs such as commissions, slippage, or spread, hence being another factor distancing the present results from potential live market testing. Still, backtests were vital for conducting the present study and remain important in the literature as a whole. Without the backtest, researchers would be limited by the duration of their live market test (a backtest trades on all available data) and by the amount of currency allotted to the algorithm, greatly prolonging the research and increasing its costs. Future research should prioritize replicating a similar strategy across a broader set of

securities and time frames, including different sectors, market capitalizations, and asset classes such as ETFs and commodities. This would determine if the present findings generalize beyond AAPL. Additionally, future research could benefit from obtaining authentic Fear & Greed data directly from CNN's index via API. Finally, replacing the strict candlestick pattern recognition system with a machine learning model trained to identify patterns statistically in the context of the market instead of with fixed thresholds may produce more adaptive entry signals, representing potential for improving upon the present strategy's performance.

Conclusion

The present study sought to determine to what extent a candlestick pattern recognition, sentiment-proxy indicator, and support and resistance algorithm provides an effective strategy for retail trading on the AAPL daily time frame. The backtest produced a net return of -40.90% and an alpha of -1260.24% against a buy and hold return of 1,219.34%, indicating the strategy failed to generate profitable returns. However, these findings suggest that the underperformance of the strategy reflects the constraints of the TradingView implementation environment rather than a fundamentally flawed theoretical framework. Each component, candlestick pattern recognition, sentiment analysis, and support resistance, has demonstrated individual effectiveness in prior research, on top of evidence that suggests a hybrid strategy could garner more profitability. The results from the present study reveal that capitalizing on those benefits require a more sophisticated coding language, development environment— such as using Python to call y*finance*— and machine learning integration than Pine Script currently offers. In this, the present findings contribute to the literature in illustrating the limitations of multi-indicator strategies for retail traders, and that a candlestick pattern recognition, sentiment-proxy indicator, and support and resistance strategy coded in TradingView's Pine Script (a popular application among retail

traders) is likely not an effective strategy for trading high performing, low volatility stocks such as AAPL. The present study suggests a clear direction for future research toward adaptive and machine learning implementations.

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